

Room-Temperature Single-Molecule Mass Detection via Feedback Control of ZnO Nanowire Resonator

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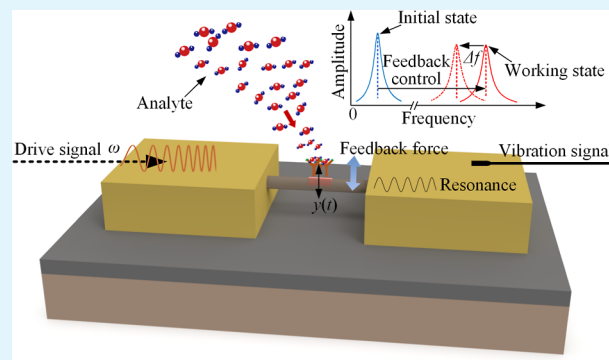
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ABSTRACT: Nanowire resonators with near single-molecule detection sensitivity are highly susceptible to environmental factors due to their small size. Typically, these sensors require ultralow temperature conditions to function effectively, which are unfavorable for monitoring biochemical reactions. To address the technical bottleneck of insufficient stability and sensitivity of nanowire resonators at room temperature, this study utilizes ZnO nanowire (NW) resonators with larger dimensions that can operate at room temperature. By employing feedback control technology, the equivalent mass of the ZnO NW resonator is significantly reduced, thereby improving its sensitivity to -2921.42 Hz/zg. This breakthrough achieves the single-molecule detection limit at room temperature, elevating the mass detection capability to the level of an ethanol molecule (approximately 0.076 zg). Furthermore, through precise deposition of a sensitive layer on the nanowire, the ZnO NW resonator enables accurate real-time monitoring of ethanol gas molecule adsorption and desorption within a vacuum chamber. The measurement error for the mass of an individual ethanol molecule is as low as 3.9%. These findings represent a substantial advancement in single-molecule detection technology.

KEYWORDS: single molecule mass detection, ZnO nanowire resonator, feedback control, room temperature, ethanol molecule



1. INTRODUCTION

Single-molecule detection technology applied to biochemical reactions holds great potential for applications in mass spectrometry, clinical diagnostics, and environmental monitoring.^{1–3} This technology can precisely capture subtle physical and chemical changes during reaction processes, providing essential data to uncover reaction mechanisms and identify early disease biomarkers. It also aids in detecting pollutants, tracing their sources, and supporting microscopic research on material structures and properties.^{4–6}

Single-molecule detection technology encompasses methodologies based on optical,⁷ electronic,⁴ and mechanical⁸ approaches. This technology is distinguished by two key features. First, it minimizes information loss caused by the collective averaging effect, enabling direct observation of molecular behavior in its native state. Second, it allows for real-time monitoring of the conformation and function of large biological molecules, thereby significantly advancing research in biochemistry and biophysics. However, optical and electronic detection methods often face limitations due to their reliance on biochemical reactions coupled with optical or electronic effects, which impose strict requirements on sample preparation and experimental techniques. Fluorescence technology has short bleaching time, low labeling efficiency, and potential protein conformational changes, which is difficult to

reflect the molecular dynamics in the natural state.⁹ Electronic methods face challenges such as demanding nanostructure fabrication precision. For example, nanopore devices require silicon nitride layer thickness errors to be below 1 nm. They also involve high equipment costs like electron-multiplying CCDs and are prone to interface effects and background noise.¹⁰ Although advancements like plasmon-enhanced spectroscopy and single-molecule junction technology have improved detection capabilities,¹¹ current optical and electronic methods struggle to detect ultralow-abundance biomarkers. They are particularly limited in detecting biomolecules with masses smaller than 10 zg. For instance, these technologies can identify proteins like α -interferon, which has a mass of approximately 33.2 zg, but they fail to detect smaller biomarkers such as RNA isomers, ultrashort-chain noncoding RNAs, one-carbon metabolism intermediates, and small peptide fragments involved in signaling.

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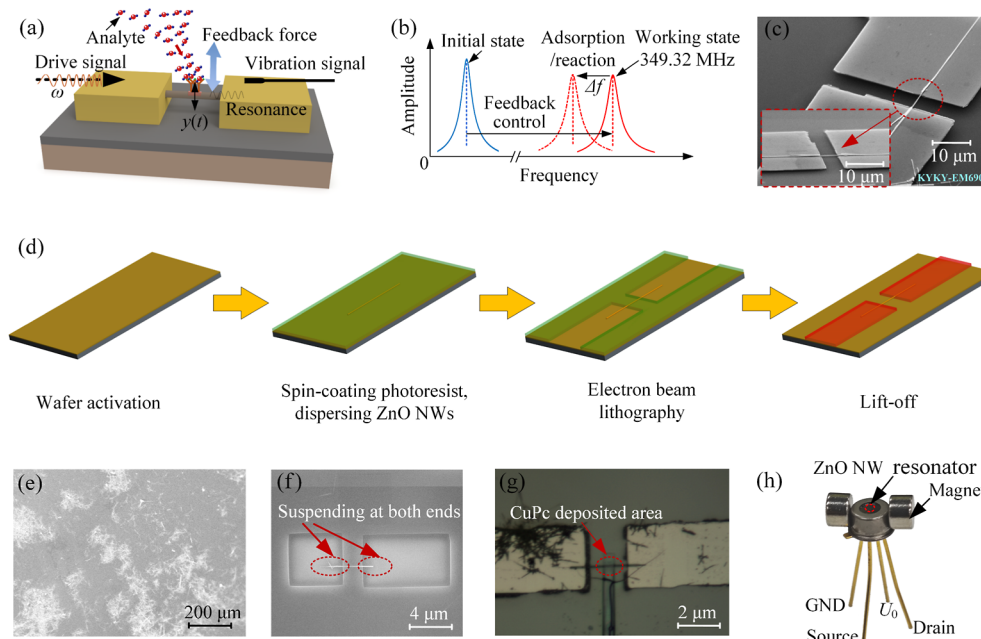


Figure 1. Fabrication process of the ZnO NW resonant sensor. (a) Schematic diagram of the single-molecule detection principle. (b) Illustration of the corresponding frequency range. (c) SEM image of the ZnO NW resonator. (d) Schematic of the resonator fabrication process. (e) SEM image of ZnO NWs dispersed on the PMMA-coated substrate. (f) SEM image showing the ZnO NW suspended at both ends after lithography. (g) Deposition of CuPc at the center of the ZnO NW. (h) Packaging process of the resonator.

Resonators provide valuable insights into intermolecular forces and reaction dynamics by monitoring changes in resonant frequencies caused by adsorbed substances on the surface, and represent a promising area of research. This technology has applications in detecting pathogen biomass, assessing cellular health status,¹² measuring the rate of cellular mass accumulation, and quantifying concentrations of cancer-related biomarkers.¹³ Additionally, it enables the study of the dynamic processes involved in the interaction between bacteria and antibodies.¹⁴ However, nanowires possess a substantial surface area, leading to a significant issue of reduced vibration stability and a low Q factor for nanowire resonators.¹⁵ The ultrasmall size of these sensors requires operation under conditions of ultralow temperature, ranging from 4 to 40 K, and under ultrahigh vacuum conditions of less than 10^{-6} Pa.¹⁶ Furthermore, most biochemical reactions heavily rely on room temperature conditions, presenting a challenge for existing nano resonant sensing technologies to satisfy the dual requirements of room-temperature operation and single-molecule sensitivity. Currently, large-sized nanowire resonators capable of operating at room temperature do not possess sufficient sensitivity to detect single molecules with a mass below 10 zg, making it challenging to identify many potential disease biomarkers.

To enhance the sensitivity of nanowire resonators based on frequency shift Δf sensing, advancing frequency tuning technology is essential. Nanowire resonators offer significant advantages in wide-range frequency tuning due to their high sensitivity to axial forces and variations in equivalent mass. Recent studies have explored an innovative feedback control technology to enhance resonator sensitivity.^{17–19} Although feedback tuning is complex to implement, it outperforms other active methods like electrothermal, electrostatic, piezoelectric, and electronic approaches such as capacitive and resistive techniques.^{20–23} Feedback tuning can achieve frequency adjustments by several orders of magnitude (up to tens of

thousands of times), whereas other tuning methods are typically limited to adjustments of a few hundred times. Furthermore, feedback tuning is based on mechanical mechanisms, offering superior precision and resistance to interference compared to electronic tuning methods. This feedback control technology optimizes the resonator's physical properties by reducing its equivalent mass or fine-tuning its equivalent damping parameters to achieve substantial performance improvements. Our preliminary work successfully achieved the sensitivity required for single-molecule detection through the feedback control technology.¹⁸ However, the relatively large equivalent mass of microcantilever resonators necessitates high feedback control gains, leading to substantial resonant frequency fluctuations that are difficult to minimize. Additionally, the lack of precise control over the distance between the gas nozzle and the resonator hampers the accurate regulation of molecular capture rates, making stable and high-precision single-molecule detection challenging.

This paper utilizes a ZnO nanowire (NW) resonator developed in our previous work^{24,25} and applies feedback control technology to enhance its sensitivity to the level required for single-molecule detection. Due to the smaller equivalent mass, it is expected that the sensitivity of the nanowire resonator can be adjusted more efficiently by feedback control. For improved test precision, the sensitive area is strategically located at the nanowire's midpoint, which standardizes the step difference of the single molecule adsorption response. Moreover, to guarantee the high-quality factor of the resonator, conducting the single-molecule mass detection experiment under a low vacuum environment is essential,^{26,27} as it ensures high detection precision. The detection mechanism of the ZnO NW resonator for analytes is illustrated in Figure 1a, with the corresponding frequency range shown in Figure 1b. Utilizing the feedback control technology enhances the sensitivity of the ZnO NW resonator, combined with precise control of analyte concentration,

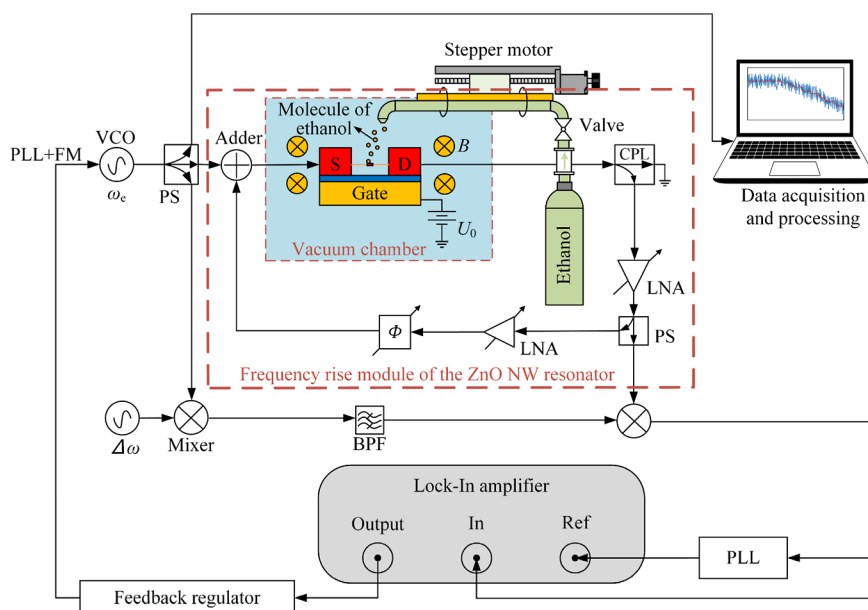


Figure 2. Sensitivity enhancement and closed-loop test scheme of the ZnO NW resonant sensor. Here, PS: power splitter, LNA: low noise amplifier, Φ : $\pi/2$ phase shifter, BPF: bandpass filter, PLL: phase-locked loop, CPL: directional coupler.

successfully enabling the detection of single ethanol gas molecule adsorption. The findings reveal that through feedback control, the sensitivity of the ZnO NW resonator can match or even surpass those of single-walled carbon nanotubes under ultrahigh vacuum and ultralow temperature conditions. This study offers novel research perspectives for mass spectrometry and atomic physics, as well as broad clinical and environmental applications.

2. MATERIAL AND METHODS

2.1. Fabrication of ZnO NW Resonator. ZnO NWs are a one-dimensional semiconductor material with well-established fabrication techniques and outstanding chemical, mechanical, and thermal stability, making them ideal for nanowire resonators. This study adopts a top-down approach to fabricate the ZnO NW resonator with fixed ends (Figure 1c). The fixed structure is successfully created using electron beam lithography and lift-off techniques, ensuring the nanowire remains undamaged by surface tension during the wet etching process. The fabrication process is illustrated in Figure 1d. A silicon wafer with a 2 μm oxide layer is used as the substrate, which is activated and spin-coated with PMMA photoresist, followed by preheating. Single-crystal ZnO NWs (diameter: 50–120 nm, aspect ratio: 60–300) are dispersed onto the photoresist surface. A secondary spin coating of 400 nm-thick PMMA is applied, and electron beam lithography is used to define electrode areas at both ends of the nanowire. Gold electrodes with thicknesses of 400–800 nm are deposited at a low rate of 0.4 $\text{\AA}/\text{s}$, and the lift-off process is performed to create the fixed structure. Finally, high-temperature annealing at 573 K reduces contact resistance and repairs defects induced by the fabrication process. This method preserves the crystalline integrity of ZnO NWs while ensuring stable device performance at room temperature.

The dispersion of ZnO NWs on the PMMA-coated substrate is shown in Figure 1e, where many ZnO NWs are unevenly dispersed on the surface. After exposing and developing the ZnO NWs, the structure shown in Figure 1f was obtained: ZnO NWs with both ends suspended while the middle section remained embedded in PMMA. To ensure the fixed support of the ZnO NWs, the thickness of the gold deposited at both ends must exceed the thickness of the initially coated PMMA layer. A layer of about 1 nm thick copper phthalocyanine (CuPc) film was precisely deposited near the middle area of the ZnO NW through the lift-off technique (Figure 1g),

followed by an annealing process (temperature set to 320 $^{\circ}\text{C}$). The aim of the annealing process was to reduce the defects within the CuPc film, while enhancing its surface adhesion to the molecules being tested. This annealing step promoted the transformation of the CuPc from the α -phase to the β -phase,²⁸ which significantly improved the film's stability in strong acid, strong alkali, and high temperature environments. Given the highly polarized electron cloud structure of CuPc molecules, they exhibit strong affinity for specific gas molecules, including ethanol, nitrogen dioxide, methanol, etc. This characteristic promotes the physical adsorption process of target molecules on the CuPc surface. It is noteworthy that CuPc exhibits a high selective adsorption capability for ethanol molecules, selectively capturing ethanol molecules in pure air.²⁹

The operation of the resonator relies on electromagnetic excitation. It utilizes a packaging technique referred to as the transistor outline (TO), and the package picture of the designed ZnO NW resonator is shown in Figure 1h. By precisely monitoring the potential changes at both ends of the ZnO NW, its vibration characteristics are indirectly captured and analyzed. Figure 2 intuitively demonstrates the closed-loop testing scheme of the ZnO NW mass sensor. A voltage-controlled oscillator (VCO) applies two voltages to the ZnO NW: one is an alternating excitation voltage U_1 , in the form of $u_1\cos(\omega_e t)$, responsible for dynamic excitation; the other is a bias voltage U_0 , expressed as u_0 , aimed at static regulation. A constant and uniform strong magnetic field is applied in the direction parallel to the substrate, with its magnetic induction intensity marked as B . The core function of the bias voltage is to generate a static Ampere force and prestretch the ZnO NW to a specific offset position. The alternating Ampere force produced by U_1 causes the resonator to vibrate periodically under the stretched state. By precisely measuring the subtle changes in the resonant frequency of the resonator, the mass of the microparticles adsorbed on its surface can be indirectly inferred and quantified, achieving high-precision mass detection and analysis.

2.2. Tuning Equivalent Mass of ZnO NW Resonator. Due to the small size of the ZnO NW resonator, the resonance signal is relatively weak. Additionally, the excitation and detection signals share the same circuit, which can easily cause loop interference. As a result, traditional feedback control methods for microcantilever resonator cannot be directly applied to the ZnO NW resonator. Therefore, it is necessary to explore which resonator response signal should be extracted, the conversion of signal phase and amplitude, and the specific impact of feedback control parameters on resonance characteristics. When the resonator is in resonant motion under the

action of U_1 , the alternating Ampere force $F_e = \frac{BU_1 \cos(\omega t)}{R} \int_0^l \phi(x) dx$ occurs,³⁰ where R , l , ω and $\phi(x)$ are the resistance, initial length, resonant frequency, mode function of the ZnO NW, respectively. That is, the transfer function between F_e and U_1 is $G_3 = Bl\epsilon^0/(2R)$. The forced vibration of the resonator exhibits an equivalent dynamic displacement, which can be mathematically represented as $y(t) = Ae^{i(\omega t - \varphi)}$, where A represents the average amplitude of the vibration.

During its vibration, the ZnO NW generates an average electromotive force given by $U_{\text{avg}}(t) = -\epsilon_l Bl \frac{dy}{dt}$, where $\epsilon_l = \frac{1}{l} \int_0^l \phi(x) dx \approx 0.5$.³¹ The electromotive force generated by the ZnO NW near its resonant frequency can be expressed as $U_{\text{avg}}(\omega, t) = \epsilon_l \frac{l^2 B^2 \omega Q U_1}{kR} e^{i\pi}$,³¹ where k denotes the stiffness that is equivalent to the resonator's characteristics. Thus, the ZnO NW resonator's output voltage can be expressed as

$$U_i = \frac{RkU_{\text{avg}}e^{i\pi}}{\epsilon_l l^2 B^2 Q \omega} = \frac{Rk}{\epsilon_l l^2 B^2 Q \omega} \epsilon_l Bl \frac{dy}{dt} = \frac{Rk}{lBQ} Ae^{i(\omega t - \varphi + \frac{\pi}{2})} = G_1 y \quad (1)$$

where $G_1 = Rke^{i\pi/2}/(lBQ)$ is the transfer function between U_i and y .

The voltage signal output during the vibration of the ZnO NW is amplified by feedback gain $G_2 = |G_2|e^{i\pi/2}$ and sent to the source of the ZnO NW as the feedback voltage U_2 . Since $d^2y/dt^2 = i^2 \omega^2 Ae^{i(\omega t - \varphi)} = \omega^2 Ae^{i(\omega t - \varphi + \pi)}$, the feedback voltage is

$$U_2 = G_2 U_i = G_1 G_2 y = |G_1||G_2| Ae^{i(\omega t - \varphi + \pi)} = \frac{|G_1||G_2|}{\omega^2} \frac{d^2 y}{dt^2} \quad (2)$$

Then F_e can be written as follows

$$\begin{aligned} F_e &= \frac{BU}{R} \int_0^l \phi(x) dx = B \frac{(U_1 + U_2)}{R} \int_0^l \phi(x) dx \\ &= \frac{BU_1}{R} \int_0^l \phi(x) dx + \frac{B|G_1||G_2|}{R} \frac{d^2 y}{dt^2} \int_0^l \phi(x) dx \end{aligned} \quad (3)$$

Based on eq 3, the model for vibration incorporating feedback control in the resonator can be described as

$$\begin{aligned} \left(m_0 - \frac{B|G_1||G_2|}{R} \int_0^l \phi(x) dx \right) \frac{d^2 y(t)}{dt^2} + \gamma_0 \frac{dy(t)}{dt} + ky(t) \\ = \frac{BU_1}{R} \int_0^l \phi(x) dx \end{aligned} \quad (4)$$

where m_0 represents the initial mass equivalent to the resonator's properties,³² γ_0 is the equivalent damping of the resonator vibration. Then the equivalent mass m_e of the resonator with feedback is $m_0 - |G|/\omega^2$, where $G = |G_1||G_2||G_3|e^{i\pi}$. Therefore, for ZnO NW resonators anchored at both ends, the resonant frequency is

$$\omega = \sqrt{\frac{k}{m_0 - |G|/\omega^2}} \quad (5)$$

Hence, precise tuning of the resonator's equivalent mass can be achieved by modifying the G value. The G_1 and G_3 are purely determined by the physical dimensions of the resonator and the bias voltage applied, and have no direct correlation with changes in excitation voltage. The values of G_1 and G_3 are determined solely by the resonator's physical attributes and the applied bias voltage, without any direct dependence on variations in the excitation voltage. By setting the phase modulation of the phase shifter to $\pi/2$ and adjusting the gain level of the low-noise amplifiers (LNAs) in the feedback circuit to modify G_2 , the quantitative change of the equivalent mass can be accurately achieved.

According to eq 2, the feedback voltage U_2 originates from the vibration response signal and is converted under the mediation of the feedback gain G . When this feedback force acts on the ZnO NW resonator, there is a π phase difference between it and the natural vibration response of the resonator. The phase difference causes the

feedback force to actually weaken the inertial effect of the resonator, thereby significantly reducing its equivalent mass. This feedback technique can be generally applied to the sensitivity adjustment of double-clamped resonators under electromagnetic excitation. Notably, this mechanism is similar to our previous research, where the mass of the microcantilever resonator was reduced using the feedback mechanism.¹⁸ However, the difference is that when the nanowire outputs a voltage signal, there is a $\pi/2$ phase offset between its equivalent vibration and the voltage signal, rather than the consistent phase situation in the microcantilever resonator.

2.3. System Stability. The stability of the system is evaluated by deeply exploring the transfer function of the system. As illustrated in Figure 3, the control diagram of the closed-loop test architecture

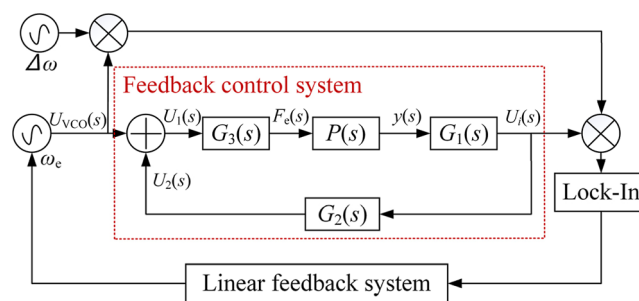


Figure 3. Control block diagram of the closed loop test system.

encompasses an internal feedback regulation mechanism in the ZnO NW resonator, aimed at decreasing its equivalent mass and improve its sensitivity. Compared to the microcantilever resonator feedback control scheme, the ZnO NW resonator requires more precise feedback gain to reduce noise. However, it operates with lower feedback gain, simplifying the circuit design. Additionally, excitation and detection signals share the same circuit, eliminating the need for a Wheatstone bridge and facilitating resonator fabrication. Nevertheless, during the generation of U_2 , a directional coupler (CPL) must be added to prevent loop interference. The entire closed-loop circuit functions in a manner akin to a closed-loop system that employs regulation based on frequency deviation.

The output voltage $U_{\text{VCO}}(t)$ of the signal generator, in conjunction with the feedback control voltage $U_2(t)$, together constitute the excitation voltage $U_e(t)$ of the resonator. This excitation voltage induces an electrostatic driving force $f(t)$ on the ZnO NW, causing the ZnO NW to vibrate and generate a displacement signal $y(t)$. Considering the resonator as a linear, time-independent system, and its transfer function explicitly describes the relationship between the excitation and the response, the specific expression is $P(s) = \frac{1}{m_v s^2 + \frac{1}{Q}s + \omega^2}$,³³ where m_v is the equivalent mass of the ZnO NW resonator.

This linear feedback system can be seen as an equivalent form of a Proportional Integral (PI) controller, and its transfer function can be rewritten as an expression like $k_p + k_i/s$, where k_i represents the integral parameter, and k_p represents the proportional parameter. Then, the open-loop transfer function of the entire control system can be expressed as

$$A(s) = G_1 G_3 P_{\text{amp}}(s) (k_p + k_i/s) \quad (6)$$

where $P_{\text{amp}}(s)$ is the transfer function between the drive signal and the output signal of the ZnO NW resonator, $P_{\text{amp}}(s) = \frac{\omega}{2k s + \frac{\omega}{2Q}}$.

Open-loop testing was performed on the fabricated ZnO NW resonator, which had a resonant length l of 6.1 μm and a nanowire diameter d of 70 nm. This was done to obtain some parameters of the ZnO NW resonator to further evaluate the stability of the closed-loop system shown in Figure 3. The first-order resonant frequency of the resonator can be estimated by $f = 0.9395d\sqrt{E}/(L^2\sqrt{\rho})$, where E is the elastic modulus of the nanowire and ρ is the density of the

nanowire.³⁴ The theoretical calculation yields a first-order resonant frequency of 10.36 MHz for this resonator (The elastic modulus of ZnO NW is approximately 2.1×10^{11} Pa, and the density is about 5.67×10^3 kg/m³).²⁵

The open-loop test for the ZnO NW resonator was conducted within a vacuum chamber (1×10^{-3} Pa) at an ambient temperature of 22 °C. To mitigate the impact of temperature variations on the resonant frequency, a water circulation system was incorporated within the vacuum chamber. This system functioned to insulate the resonator that was in contact with the chamber, facilitating heat transfer. Consequently, the surface temperature of the nanowire was effectively sustained within a narrow range of 21.0 to 21.2 °C, thus ensuring the accuracy and precision of the measurements. To further mitigate the reduction in resonant frequency, frequency fluctuations, and to enhance the *Q* factor of the resonator caused by the adsorption of gas molecules, the resonator was kept inside a vacuum chamber for over 12 h. This treatment effectively reduced the adsorption of impurities on the ZnO NW, ensuring the accuracy and reliability of single-molecule detection. Throughout the measurement process, a sweep drive voltage was set with an amplitude of 5 V. Additionally, the surrounding magnetic field strength of the resonator was set at 0.5 T, and the bias voltage amplitude was established at 2 V.

Through open-loop measurement, the resonator's first-order resonant frequency was determined to be 10.352 MHz, with a *Q* value of approximately 600. These findings are presented in Table 1. The resonant frequency measured align well with the calculated data, suggesting that the resonant frequency measured corresponds to the bending vibration of the ZnO NW.

Table 1. Test Results of the Resonator Parameters under Initial Conditions

$ U_1 $ (V)	$ U_1 $ (mV)	R (k Ω)	$ G_3 $ (N/V)	$ G_1 $ (V/m)	k
5	3.13	400	3.81×10^{-12}	6.34×10^7	0.29

To achieve a balance between noise levels and bandwidth, the integral parameter k_i is set to 9.56×10^8 . To ensure that the zero point of the gain $A(s)$ is at the resonant frequency, the proportional parameter k_p is set to 2400. The transfer function for the entire control system in open-loop is

$$A(s) = G_1 G_3 P_{\text{amp}}(k_p + k_i/s) = \frac{6.5 \times 10^7 s + 2.59 \times 10^{13}}{s^2 + 5.42 \times 10^4 s} \quad (7)$$

Based on the Bode Plot (Figure 4) drawn from eq 6, the plot reveals that at the resonant frequency of 10.352 MHz, the system gain

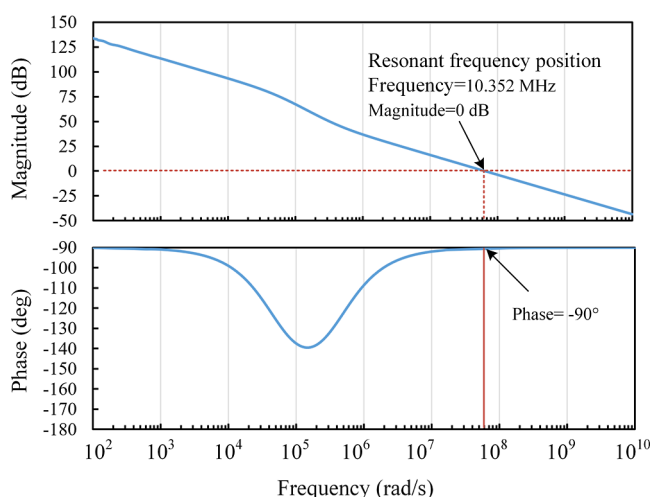


Figure 4. Bode diagram corresponding to the open-loop transfer function $A(s)$ of the whole control system.

is 0 dB, and the phase margin is 90°. To ensure system stability, the phase margin usually needs to be greater than 30°, so it can be judged that the system is stable at the resonant frequency.

The resonant frequency closed-loop detection is realized by adopting phase-locked loop (PLL) and frequency modulation (FM) techniques.²⁴ The amplitude-frequency characteristic curve, derived from frequency sweep analysis, facilitated the fine-tuning of the VCO's driving voltage. This adjustment aimed to circumvent the resonator's nonlinear behavior and prevent saturation of the feedback control components. Additionally, the incorporation of a fine-tuning phase shifter with a lock-in amplifier (OE1022, Sine Scientific Instrument Co., Ltd.) enabled precise locking of the resonant frequency. The spectrum analyzer was employed to record the output frequency of the VCO, while the frequency standard deviation f_{RMS} was computed to gauge the stability of the resonant frequency.

3. RESULTS AND DISCUSSION

3.1. Tuning Resonant Frequencies. To enhance the identification ability of the resonators to ethanol molecules, it is necessary to adjust their equivalent mass parameters with the aid of a feedback control mechanism. In this study, the phase-locked loop combined with frequency modulation was used to perform closed-loop testing of the ZnO NW resonator (see Supporting Information). The results of the closed-loop feedback test are shown in Figure 5. As shown in Figure 5a, as the feedback gain increases, the resonant frequency of the ZnO NW resonator gradually increases. The measured values and calculated values of the resonant frequency coincide, which proves the correctness of the theoretical model. The sensitivity of the resonator can be represented as $S = -f/(2m_v)$.^{35,36} When $|G_2| = 87945$, the resonant frequency increases to 349.324 MHz, with a deviation of only 1.32% from the theoretical value. Meanwhile, the sensitivity reaches -2921.42 Hz/zg. As shown in Figure 5b, as the feedback gain increases, the frequency fluctuations gradually increase, and when $|G_2| = 87945$, $f_{\text{RMS}} = 2395$ Hz. This is due to the fact that as the equivalent mass decreases, the impact of energy dissipation on the vibration becomes greater and greater. However, as the feedback gain increases, f_{RMS}/f gradually decreases, indicating that as the feedback gain increases, the sensitivity of the resonator gradually improves.

As shown in Figure 5c, as the feedback gain gradually increases, the *Q* factor shows a gradual declining trend, but this rate of decline gradually slows down. In order to accurately describe and predict the changes in the *Q* factor, the formula $Q = m_e \omega / \gamma_0$ can be used for modeling,³⁷ where the damping coefficient corresponding to the resonant frequency of 130.361 MHz is set to 5.86×10^{-12} N·s/m, as the inherent damping characteristic of the resonator. Based on this setting, the theoretical calculation curve of the impact of feedback gain on the *Q* factor is plotted. It is found that the curve closely aligns with the measured *Q* value data points, validating the accuracy of the proposed model. Furthermore, it confirms that the feedback control method utilized in this work functions by reducing the resonator's equivalent mass through feedback gain, rather than increasing its equivalent stiffness. When the resonant frequency increased to 349.324 MHz, the measured *Q* factor dropped to 40. The significant fluctuation brought the resonator close to the limit of closed-loop locking. If further increases in the resonant frequency through feedback gain are required, it is necessary to optimize the resonator's design and improve the working environment to enhance the *Q* factor.

The rule of the feedback voltage amplitude u_2 changing with the resonant frequency is shown in Figure 5d, and the straight

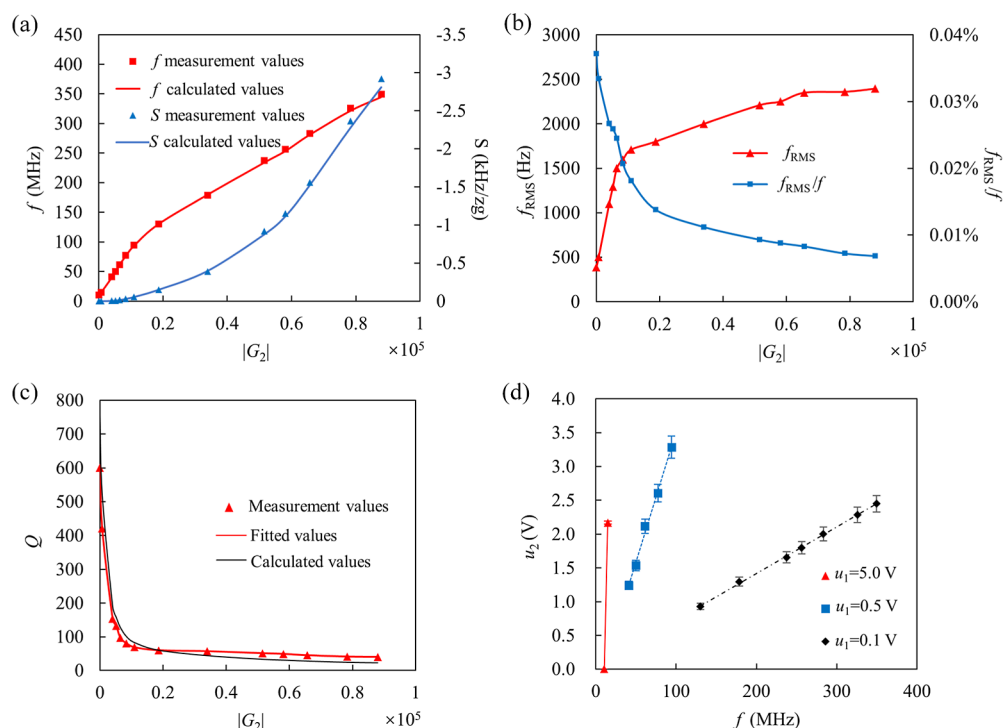


Figure 5. Results of the feedback control tests on the ZnO NW resonator. (a) Effect of feedback gain on resonant frequency and sensitivity. (b) Effect of feedback gain on f_{RMS} . (c) Effect of feedback gain on the Q value. (d) The relationship of the feedback voltage with the change in resonant frequency.

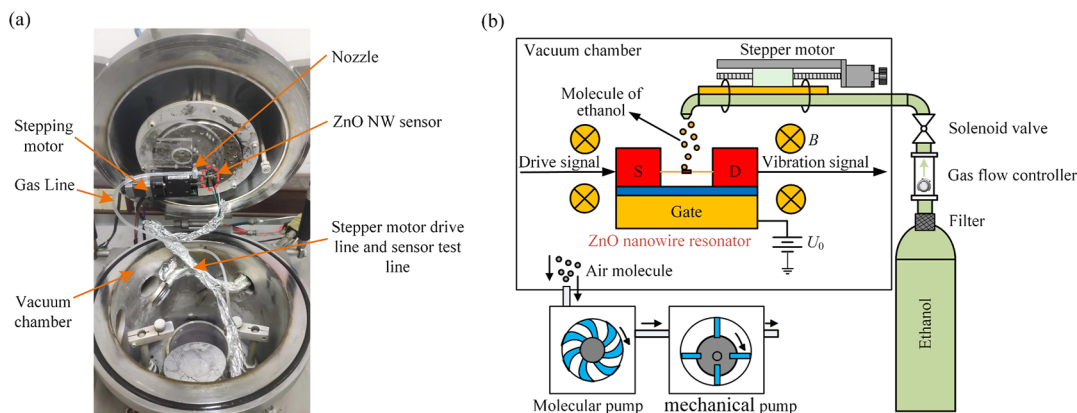


Figure 6. Ethanol gas flow control device. (a) Arrangement of the test device in the vacuum chamber. (b) Schematic diagram of ethanol gas flow control.

line in the figure is the fitting value of the measured value. For the ZnO NW resonator, according to eq 1, the amplitude of the output voltage of the resonator is

$$u_i = |G_1|A = \frac{Rk_{re}}{lBQ}A \quad (8)$$

Based on eqs 5 and 8, the amplitude of the feedback voltage can be derived as

$$u_2 = |G_2|u_i = |G_2||G_1|A = \frac{|G|}{|G_3|}A = \alpha \frac{2R}{Blf}[(2\pi f)^2 m_0 - k] \quad (9)$$

where $\alpha = Af$ represents the scale factor, which is related to the size parameters of the resonator and the excitation parameters, and can be determined by measuring the amplitude of the feedback voltage.

When f is greater than 50 MHz, the first term in eq 9 is far greater than the second term, and u_2 is approximately linearly related to f . As shown in Figure 5d, when the amplitude of the excitation voltage remains constant, the measured fitting curve of u_2 and f approximates a linear relationship. The deviation between the average measured value of u_2 and the fitted value is less than 5%, which confirms the accuracy of the feedback control model.

3.2. Single Ethanol Molecule Detection. Ethanol molecules were detected in a vacuum environment (1×10^{-3} Pa), and the arrangement of the test device in the vacuum chamber is shown in Figure 6a. A precise electromagnetic valve and gas flow control device are used to control the gas flow rate. The ethanol gas flow control is depicted in Figure 6b. To minimize the potential impact of water molecules and other gas components on the test outcomes, nitrogen gas was

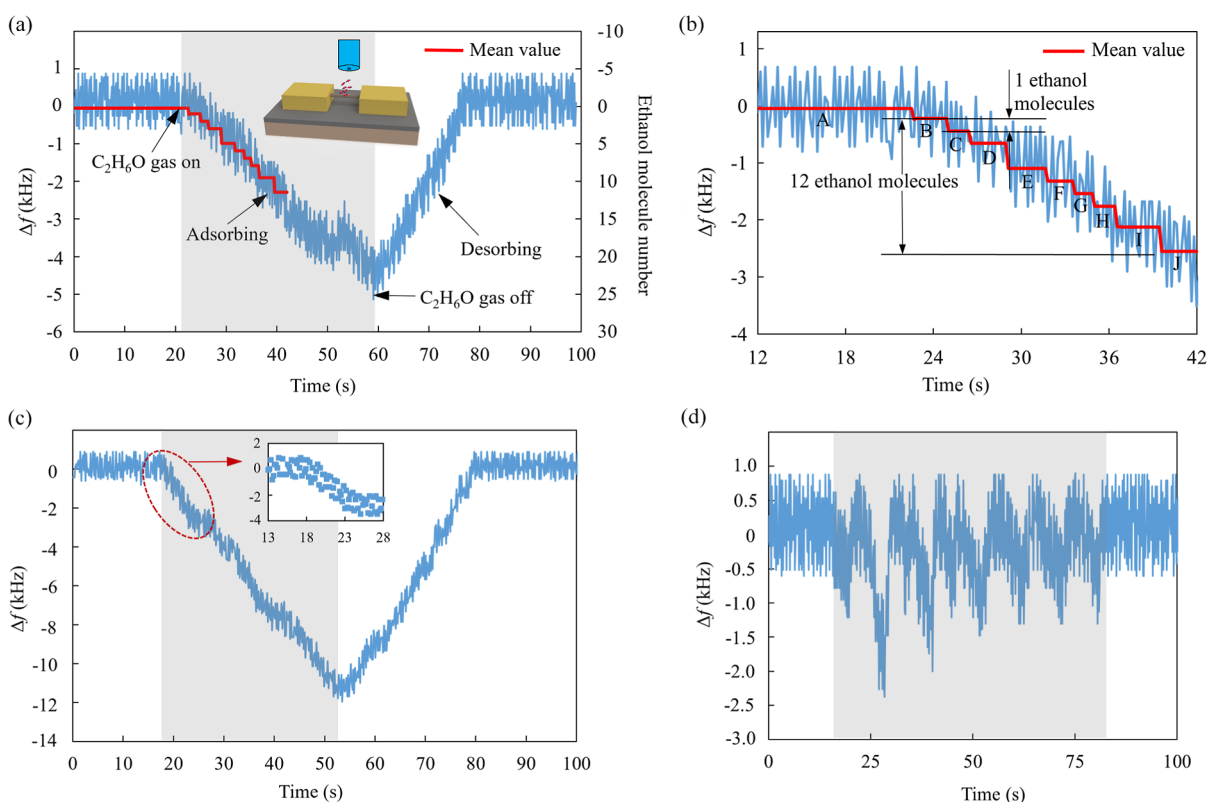


Figure 7. Response of the ZnO NW resonator to ethanol gas. (a) Response of the ZnO NW resonator to ethanol gas when $l_{sd} = 10$ mm, the inset represents a schematic diagram of the diffusion of ethanol molecules toward the ZnO NW resonator, and (b) response curve corresponding to 12 to 42 s. (c) Response of the ZnO NW resonator to ethanol gas when $l_{sd} = 5$ mm. The inset shows the response curve for 13 to 28 s. (d) Response of the ZnO NW resonator to ethanol gas when $l_{sd} = 20$ mm. The shaded part in the figures indicates that the solenoid valve is open, and the blank part indicates that the solenoid valve is closed.

initially introduced into the vacuum chamber as a carrier gas. In the experimental process, an abundant supply of high-purity nitrogen gas (99.999%) was initially injected into the ethanol evaporation chamber to displace the original gases present. Utilizing a gas flow controller and a solenoid valve, high-purity ethanol gas ($\geq 99.9\%$) was accurately directed toward the vicinity of the ZnO NW resonator through a fine tube outfitted with a filter. These comprehensive measures effectively guaranteed the purity of the environment within the vacuum chamber. The gas flow controller includes a digital gas flow meter (CS200C, Beijing Sevenstar Flow Co., Ltd.) with a capacity of 2 sccm and a mass flow controller (D08-2E, Beijing Sevenstar Flow Co., Ltd.). The distance between the gas nozzle and the resonator is precisely controlled by adjusting the stepper motor, thereby controlling the sensor's adsorption rate of ethanol gas within an observable range.

At room temperature (22 °C), the gas nozzle was set to a distance l_{sd} of 10 mm from the ZnO NW resonator, enabling the resonant frequency to stabilize at approximately 349.324 MHz. Following this, the solenoid valve was opened, and the mass flow controller reduced the ethanol flow to a minimal level, with the potentiometer adjusted to a 1% opening. Given that mechanical and molecular pumps were continuously extracting gas molecules from the vacuum chamber, the trace amounts of ethanol gas introduced into the resonator did not significantly increase the pressure within the vacuum chamber. Furthermore, once the gas flow was shut off, the ethanol gas introduced into the vacuum chamber was rapidly removed, leading to the desorption of ethanol molecules previously

adsorbed onto CuPc due to the decreased concentration of ethanol molecules.¹⁷

High-speed photography was employed to capture the spectrum of the VCO output signal presented by the spectrum analyzer at a rate of 60 frames per second. To rapidly analyze changes in the resonant frequency, data was collected using the cloud image recognition software provided by iFlytek Co., Ltd. During the experiment, the solenoid valve was operational from approximately the 22 s mark until around 60 s. The test results are presented in Figure 7a. The resonant frequency's drop and rise phases corresponded with the gas flow switch time points, indicating that the frequency shifts were directly triggered by the adsorption and desorption of CuPc to ethanol.

The analysis of the frequency change curve provides insights into the resonance behavior characteristics of the ZnO NW resonator during the adsorption of ethanol molecules. As illustrated in Figure 7b, the resonant frequency experienced a decline of 2544 Hz from stage A to stage J throughout an approximately 22 s adsorption cycle. The Δf triggered by a single ethanol molecule can be determined by multiplying the sensitivity and the mass of one ethanol molecule.^{35,36} This indicates that from stage A to stage J corresponds to CuPc capturing about 12 ethanol molecules. In the same way, it can be known that 1,1,1 and 2 ethanol molecules are respectively adsorbed on the resonator in stages A-B, B-C, C-D, and D-E. From stage A to stage J, the average frequency shift Δf_r induced by the adsorbed single ethanol molecules is 212 Hz. Consequently, the measured mass of a single ethanol molecule Δm_{pt} can be calculated as $\Delta f_r/S = 0.073$ zg. Given that the

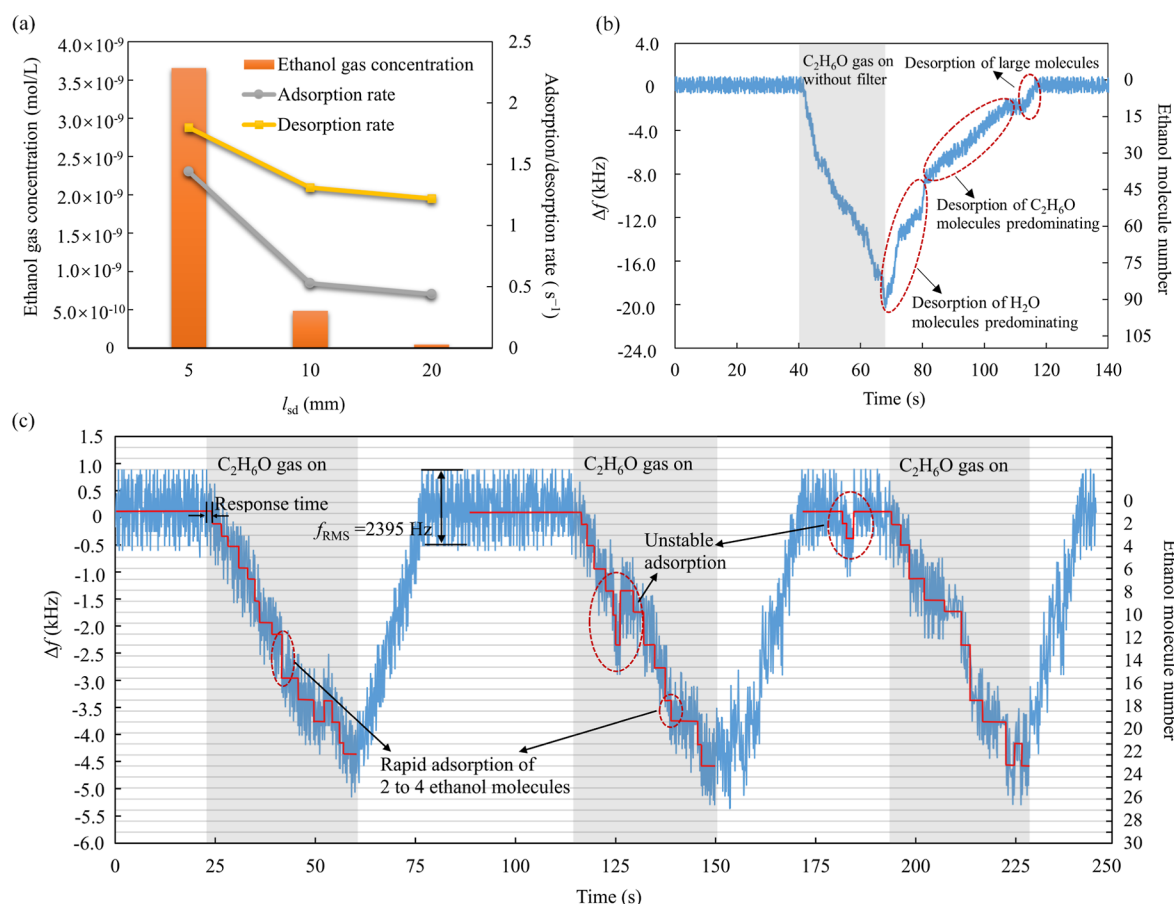


Figure 8. (a) Effect of concentration of ethanol gas on adsorption and desorption rate. (b) Response of the ZnO NW resonator without a filter. (c) Three-time repeated response of the ZnO NW resonator to ethanol gas when $l_{sd} = 10$ mm.

standard mass of an ethanol molecule is approximately 0.076 zg (molecular weight of 46), the relative error between the measured value and the theoretical value for the mass of a single ethanol molecule is 3.9%. The primary sources of this error are thermal noise from the ZnO NW resonator and fluctuations in the resonant frequency caused by environmental factors. Averaging the frequency shifts induced by multiple ethanol molecules can effectively reduce the measurement error. The stepping on the resonant frequency–time curve can be attributed to the adsorption of single ethanol molecules, rather than the fluctuation of the resonant frequency, thereby achieving the level of single molecule recognition. Although the detected root-mean-square frequency fluctuation f_{RMS} is as high as 2395 Hz, far exceeding the frequency change caused by the adsorption of a single molecule, the subtle differences in frequency fluctuations can still effectively identify the adsorption events of CuPc to ethanol molecules. This characteristic significantly expands the potential applications of the resonator and enhances its ability to detect minute mass changes. Although CuPc also exhibits affinity for other gases, such as methanol (CH₃O), the mass detection precision of 3.9% enables the resonator to distinguish the adsorption and desorption behaviors of methanol and ethanol in mixed gases. This differentiation is achieved by leveraging the significant mass difference between the two gases (with CH₃O as the reference, the mass difference is 43.8%) and the resulting distinct variations in the Δf . Such capability provides technical support for the simultaneous

detection of multiple analytes, thereby enhancing the practical value and application potential of the resonator.

In comparison to the reaction of microcantilever resonators to ethanol molecules within an atmospheric environment,¹⁸ the ZnO NW resonator exhibits a more stable response speed in a low vacuum environment. Likewise, the desorption response of CuPc to ethanol molecules is more stable than its adsorption. Upon complete desorption of ethanol molecules from the resonator, the resonant frequency stabilizes, approximating the initial resonant frequency. This suggests that the recovery performance of the ZnO NW resonator is commendable. In the course of the experiment, the relative humidity outside the vacuum chamber was consistently held between 45% and 50%. The resonant frequency remained stable at the baseline value of 349.324 MHz without the introduction of the test gas, even as the relative humidity rose from 45% to 50%. Concurrently, the f_{RMS} stabilized at 2395 Hz. These findings suggest that the impact of humidity fluctuations on resonant frequency is not as obvious as that of the inherent resonant noise.

The concentration of ethanol gas has great influence on the resonant frequency–time curve. In order to study this influence, the l_{sd} was set to 5 mm and 20 mm, respectively. The test results of the effect of this adjustment on the reaction characteristics of the ZnO NW resonator are shown in Figure 7c,d, respectively. When l_{sd} is set to 5 mm, compared to $l_{sd} = 10$ mm, the adsorption and desorption speed of ethanol molecules is significantly increased, and the decrease process of the resonant frequency hardly shows any obvious signs of

recovery. This phenomenon is attributed to the shortened distance between the nozzle and the resonator, keeping the concentration of ethanol gas around the ZnO NW resonator at a relatively constant level. When l_{sd} is set to 20 mm, the resonant frequency changes slightly. However, this decrease in frequency does not sustain, suggesting that the ethanol molecules have evaporated close to the nanowire's surface and have been adsorbed by CuPc, but due to the lack of a constant ethanol gas concentration, the adsorbed ethanol molecules do not stay on the CuPc surface for an extended period before desorption. The uneven and sparse presence of ethanol molecules leads to the irregular adsorption and desorption of ethanol molecules by CuPc.

The ethanol gas response curve for the ZnO NW resonator corresponding to different l_{sd} was used to calculate the adsorption and desorption rates of ethanol molecules. As shown in Figure 8a, the effect of l_{sd} on the resonator was primarily reflected in the adsorption rate. When l_{sd} was set to 20 mm, the molecular adsorption rate decreased from 1.44 s^{-1} (at $l_{sd} = 5\text{ mm}$) to 0.44 s^{-1} , representing a decline of approximately 69.4%. In contrast, the molecular desorption rate decreased from 1.80 s^{-1} to 1.22 s^{-1} , which corresponds to a reduction of only about 32.2%. This indicates that precise control of the distance between the analyte nozzle and the resonator is crucial for experiments aimed at analyzing biochemical reaction kinetics by observing molecular adsorption phenomena. The ethanol concentration surrounding the resonator was determined by measuring the pressure using an ionization gauge. Positioned approximately l_{sd} from the nozzle, the ionization gauge showed that the surrounding pressure p rapidly increased due to the injection of ethanol gas. Considering the atmospheric pressure p_a of $1.0 \times 10^5\text{ Pa}$, which corresponds to a nitrogen concentration n_n of 0.045 mol/L, the ethanol gas concentration n can be calculated based on p relative to the initial vacuum chamber pressure p_v of $1.0 \times 10^{-3}\text{ Pa}$

$$n = \frac{n_n(p - p_v)}{p_a} \quad (10)$$

For l_{sd} values of 5 mm, 10 mm, and 20 mm, the pressures measured by the ionization gauge were $9.3 \times 10^{-3}\text{ Pa}$, $2.1 \times 10^{-3}\text{ Pa}$, and $1.1 \times 10^{-3}\text{ Pa}$, respectively. The corresponding ethanol gas concentrations shown in Figure 8a were calculated to be $3.7 \times 10^{-9}\text{ mol/L}$, $4.9 \times 10^{-10}\text{ mol/L}$, and $4.5 \times 10^{-11}\text{ mol/L}$. Using this method, a relationship between the analyte concentration and biochemical reactions can be established. The inability to sustain resonant frequency decreases for $l_{sd} = 20\text{ mm}$ indicates that $4.5 \times 10^{-11}\text{ mol/L}$ represents the lowest concentration at which single-molecule detection can be achieved for this resonator. This may be attributed to the relatively small size of the NW resonator's sensitive layer, resulting in a detection limit higher than the sample concentrations typically required for single-molecule detection, which range from 10^{-13} to 10^{-17} mol/L .^{7,38}

The impact of setting the filter was analyzed. As shown in Figure 8b, the test results without filter show that the decrease in resonant frequency is significantly accelerated. This is due to the simultaneous adsorption of some water and ethanol molecules by the CuPc film. When the solenoid valve is closed, the recovery speed of the resonant frequency exhibits a trend of first fast then slow, which is because the desorption speed of water molecules is faster than that of ethanol molecules; that is,

after the water molecules are completely desorbed, there still exist some ethanol molecules that have not desorbed. A rapid rise appears in the final stage of the resonant frequency recovery, which is because without a filter, it is easier to introduce some large molecules that are adsorbed onto the CuPc film, and these large molecules are generally difficult to desorb.

As a sensor that relies primarily on changes in resonator frequency steps to detect mass fluctuations for single-molecule detection, the impact of long-term material stability changes is smaller compared to sensors that rely on electrical characteristics such as resistance changes. As shown in Figure 8c, the response and recovery speeds of the ZnO NW resonator remained consistent across three repeated experiments, without any observed performance degradation or overall response shift of the device. Although instabilities in gas molecule adsorption occurred once during the second and third experiments, these fluctuations fall within the normal range for trace gas adsorption. The molecular adsorption rates in the three repeated experiments were 0.53 s^{-1} , 0.55 s^{-1} , and 0.55 s^{-1} , with a deviation of only 3.8%. Despite a Q factor of only 40, frequency fluctuations did not increase during the three repeated experiments. Furthermore, complete desorption of ethanol molecules was achieved in multiple experiments, demonstrating the stability and reproducibility of the resonator. The time from the introduction of the target molecules to the resonant frequency exhibiting a step change was approximately 1.8 s. To reduce measurement error, the step response time for several frequency steps was about 20 s. However, the molecular adsorption onto CuPc occurred within nanoseconds, and the main time delay originated from the transmission of molecules from the nozzle to the resonator. Compared to advanced optical-based single-molecule detection methods, which require tens of seconds for overall detection,³⁹ this approach significantly improves detection precision while maintaining comparable overall detection efficiency. Moreover, compared to traditional single-molecule detection methods that require several minutes, the overall detection efficiency has been greatly improved. This breakthrough marks dual progress in both efficiency and precision for single-molecule detection technology. Future research can focus on optimizing the testing environment and device structure to improve the Q factor, further reducing frequency fluctuations. This enhancement would allow for higher resonant frequency and sensitivity, while decreasing the number of frequency steps required for measurement. Ultimately, this could enable rapid detection of all small molecules, including hydrogen molecules.

The experimental analysis presented above reveals a relatively high initial resonant frequency for the ZnO NW resonator. This allows for substantial sensitivity without necessitating a large feedback gain. Compared to other nanowire resonators, the ZnO NW resonator provides distinct advantages including the potential for higher sensitivity, streamlined testing procedures, and cost reductions, as depicted in Table S1. Although the structure of the CuPc coating may change over time, this only affects the adsorption and desorption rates of the resonator. It does not affect the accuracy of detection method based on resonant frequency shifts caused by mass changes. For each operation, molecular detection with distinct step differences can be achieved by simply fine-tuning the stepper motor to adjust to the appropriate analyte concentration. Future research can focus

on designing nanowire resonators with diameters exceeding 500 nm to enhance interference resistance, enabling operation in atmospheric environments. Furthermore, the integration of feedback frequency modulation techniques can improve detection sensitivity and extend the detection range, thereby achieving the unification of high sensitivity, broad detection range, and strong environmental adaptability. The ZnO NW sensor can be employed to detect various types of molecules, enabling the exploration of dynamic mechanisms underlying biochemical processes, such as nanoparticle–gas interactions, antigen–antibody binding, single-atom catalysis, and viral replication. For atoms or molecules, such as metals, alloys, and compounds, physical vapor deposition (PVD) can be employed to transfer the target material onto the resonator. Gas molecules can be introduced to the resonator using vacuum chamber gas injection techniques. For solid-phase or liquid-phase biological particles, including antigens, antibodies, viruses, enzymes, and RNA, sample injection can be achieved using matrix-assisted laser desorption/ionization (MALDI) or high-precision electrospray ionization (ESI).^{40,41}

4. CONCLUSION

This study presents the design of a high-sensitivity ZnO NW resonator, utilizing a feedback control method. The feedback control dynamic model for this resonant mass sensor is proposed and established, providing insight into the influence of feedback control parameters on resonance characteristics. This resonator was employed for real-time detection of adsorbed ethanol gas molecules under low vacuum conditions. The designed closed-loop test system has the ability to detect and feedback the weak signals. Feedback technology serves to modulate the frequency of the ZnO NW resonator, thereby enhancing its sensitivity. Through feedback control, the resonant frequency has been successfully increased from 10.36 to 349.324 MHz, and the mass detection sensitivity has reached -2921.42 Hz/zg. The designed ZnO NW resonator has a mass resolution with single molecule accuracy at room temperature.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author on reasonable request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c06688>.

Closed-loop control system of ZnO NW resonator and comparison of nanowire resonators in this paper with resonators described in other literature (PDF)

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X.C.: Writing—review and editing, Writing—original draft, Methodology, Investigation, Formal analysis, Data curation. X.W., Z.F., Y.C., and X.Z.: Validation, Methodology, Investigation, Formal analysis, Data curation. W.L.: Writing—review and editing, Methodology, Project administration, Funding acquisition. All authors have given approval to the final version of the manuscript.

Notes

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